Ex.No.2 - Geometric Primitives - Transformations

- *Name: Monish D.Y.*
- *Roll No.: 230701195*
- *Date: 10/01/2026*

In [7]:
```
import numpy as np
import cv2 as cv
import matplotlib.pyplot as plt
```

In [236…
```
# Read an image from the path using OpenCV
path = r"D:\VI\IP & CV\Lab Experiments\Inputs\vijay.jpg"
img = cv.imread(path)

# Show the image
cv.imshow('Vijay', img)
cv.waitKey(0)
cv.destroyAllWindows()
```

In [237…
```
# Resizing the image to its half the width and half the height using resize()
res = cv.resize(img, None, fx=0.5, fy=0.5, interpolation = cv.INTER_CUBIC)

# Show the image
cv.imshow('Resized Image', res)
cv.waitKey(0)
cv.destroyAllWindows()
```

In [238…
```
# Resizing the image to its half the width and half the height using resize()
h, w = img.shape[:2]
img = cv.resize(img,(w//2, h//2), interpolation = cv.INTER_CUBIC)

# Show the image
cv.imshow('Resized Image', img)
cv.waitKey(0)
cv.destroyAllWindows()
```

In [239…
```
# Translate the image 100px right and 50px down
rows,cols,width = img.shape
M = np.float32([[1,0,100],[0,1,50]])
print(M)

# To get full image add the translated value with row and col
dst = cv.warpAffine(img,M,(cols + 100,rows + 50))

# Show the image
cv.imshow('Translated Image', dst)
cv.waitKey(0)
cv.destroyAllWindows()
```

```
[[  1.   0. 100.]
 [  0.   1.  50.]]
```

In [72]:

```
# Rotate the image to 90 degrees
M = cv.getRotationMatrix2D(((cols-1)/2.0,(rows-1)/2.0),90,1)
print(M)
dst = cv.warpAffine(img,M,(cols,rows))

# Show the image
cv.imshow('Rotated Image', dst)
cv.waitKey(0)
cv.destroyAllWindows()
```

```
[[ 6.12323400e-17  1.00000000e+00 -2.84217094e-14]
 [-1.00000000e+00  6.12323400e-17  3.99000000e+02]]
```

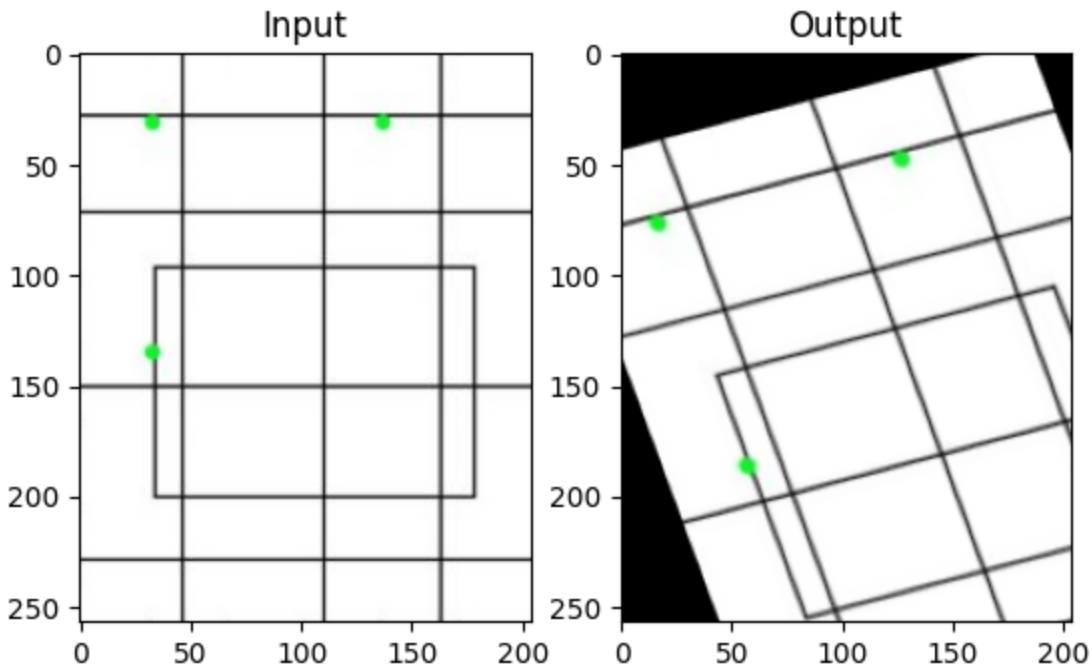
In [240…

```
# Affine Transformation
path2 = r"D:\VI\IP & CV\Lab Experiments\Inputs\drawing.pnG"
img2 = cv.imread(path2)
rows2,cols2,width2 = img2.shape

#Selecting two points
pts1 = np.float32([[20,50],[200,50],[50,200]])
pts2 = np.float32([[10,100],[200,50],[100,250]])

# Making the Affine Transformation Matrix
M = cv.getAffineTransform(pts1,pts2)
print(M)
```

```
[[  1.05555556   0.38888889 -30.55555556]
 [ -0.27777778   1.05555556  52.77777778]]
```

In [241…

```
# Affine Transformed Image
dst = cv.warpAffine(img2,M,(cols2,rows2))

# Show as Subplots
plt.subplot(121),plt.imshow(img2),plt.title('Input')
plt.subplot(122),plt.imshow(dst),plt.title('Output')
plt.show()
```



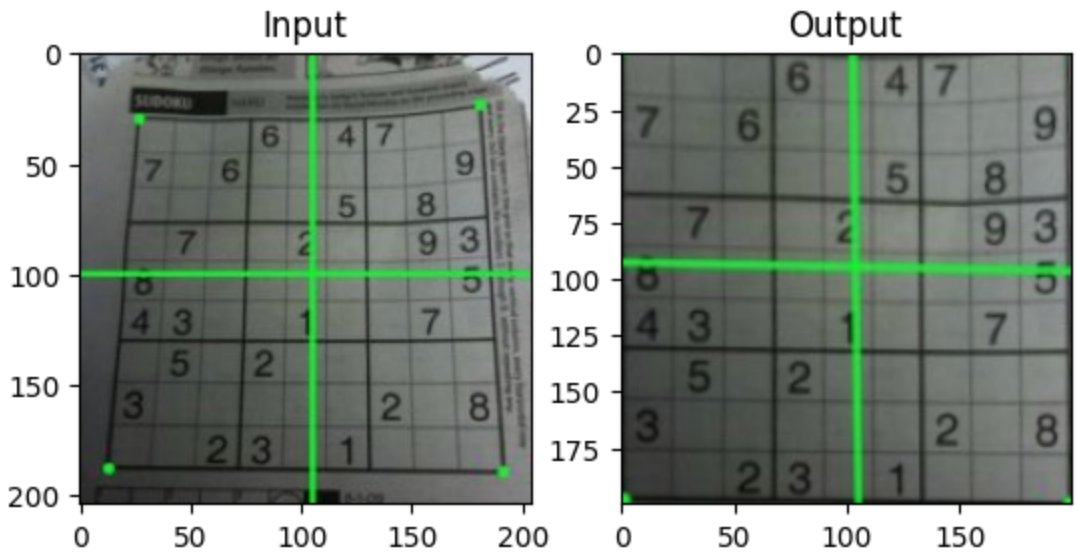
In [243…

```
path3 = r"D:\VI\IP & CV\Lab Experiments\Inputs\sudoku.png"
img3 = cv.imread(path3)

rows3,cols3,ch3 = img3.shape
pts1 = np.float32([[26,32],[183,27],[12,190],[193,190]])
pts2 = np.float32([[0,0],[200,0],[0,200],[200,200]])

M = cv.getPerspectiveTransform(pts1,pts2)
print(M)
```

```
[[ 1.36885430e+00  1.21290887e-01 -3.94715202e+01]
 [ 4.83137608e-02  1.51705209e+00 -4.98018247e+01]
 [ 2.41568804e-04  1.01152822e-03  1.00000000e+00]]
```

In [244…

```
dst = cv.warpPerspective(img3,M,(200,200))
plt.subplot(121),plt.imshow(img3),plt.title('Input')
plt.subplot(122),plt.imshow(dst),plt.title('Output')
plt.show()
```



In [247…

```
# Saving the image
cv.imwrite(r"D:\VI\IP & CV\Lab Experiments\Outputs\sudoku_warped.jpg",dst)
```

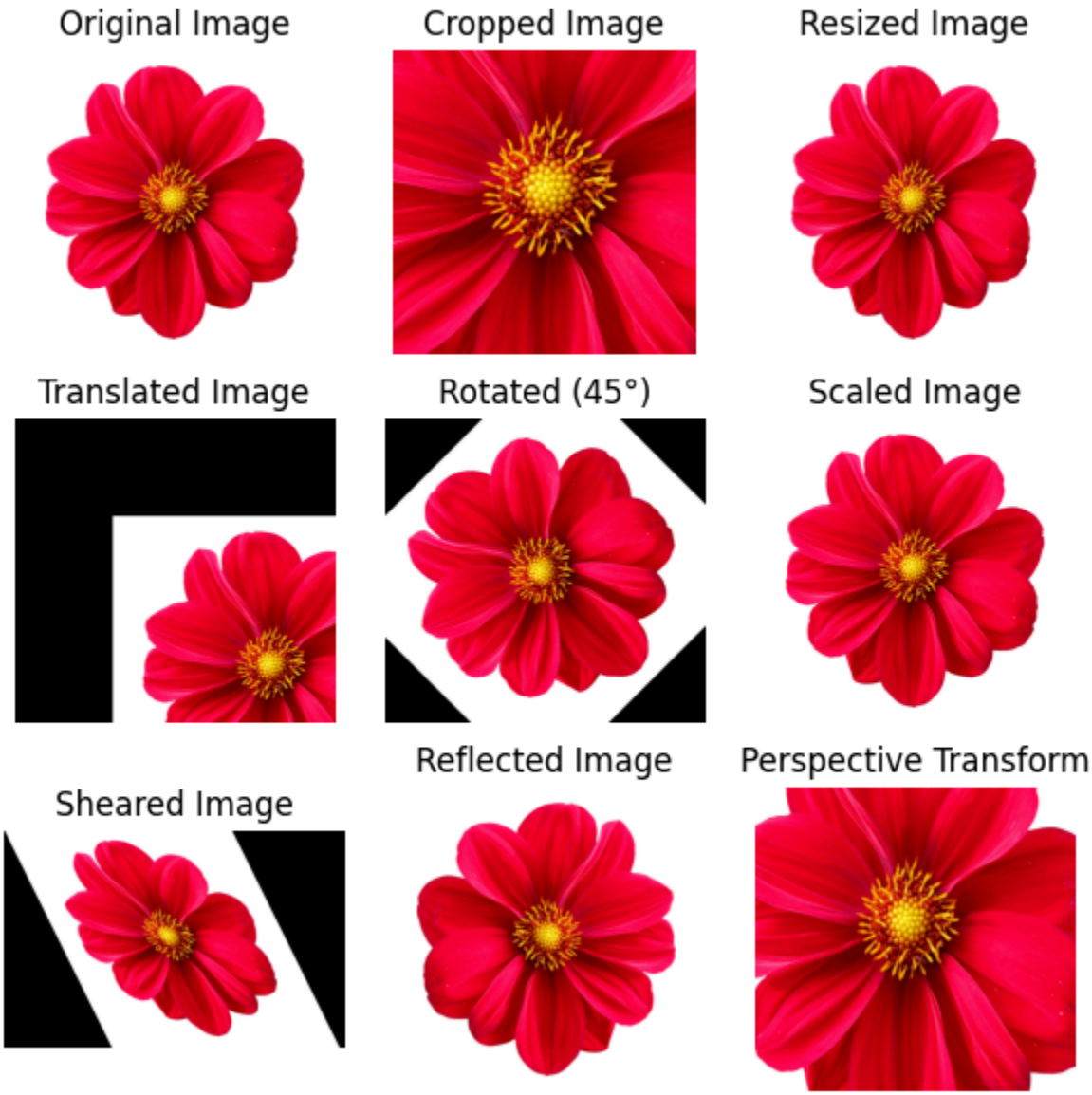
Out[247…

```
True
```

In [249…

```
# Read image and convert from BGR to RGB for Matplotlib
path4 = r"D:\VI\IP & CV\Lab Experiments\Inputs\flower.png"
img4 = cv.imread(path4)
img_rgb = cv.cvtColor(img4, cv.COLOR_BGR2RGB)

# Get image dimensions
rows4, cols4 = img4.shape[:2]

# Create a 3x3 figure layout
plt.figure("Image Transformations", figsize=(6, 6))

# ---------------------------------------------------
# 1. Original Image
# ---------------------------------------------------
plt.subplot(3, 3, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

# ---------------------------------------------------
# 2. Cropped Image
# ---------------------------------------------------
# Crop a central region of the image
cropped = img4[400:1200, 400:1200]
cropped_rgb = cv.cvtColor(cropped, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 2)
plt.imshow(cropped_rgb)
plt.title("Cropped Image")
plt.axis("off")
```

```python
# ---------------------------------------------------
# 3. Resized Image
# ---------------------------------------------------
# Resize image to fixed dimensions
resized = cv.resize(img4, (500, 500))
resized_rgb = cv.cvtColor(resized, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 3)
plt.imshow(resized_rgb)
plt.title("Resized Image")
plt.axis("off")

# ---------------------------------------------------
# 4. Translated Image
# ---------------------------------------------------
# Translate image by 500 pixels in x and y directions
M_trans = np.float32([[1, 0, 500],
                      [0, 1, 500]])
translated = cv.warpAffine(img4, M_trans, (cols4, rows4))
translated_rgb = cv.cvtColor(translated, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 4)
plt.imshow(translated_rgb)
plt.title("Translated Image")
plt.axis("off")

# ---------------------------------------------------
# 5. Rotated Image
# ---------------------------------------------------
# Rotate image by 45 degrees about its center
center = (cols4 // 2, rows4 // 2)
M_rot = cv.getRotationMatrix2D(center, 45, 1.0)
rotated = cv.warpAffine(img4, M_rot, (cols4, rows4))
rotated_rgb = cv.cvtColor(rotated, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 5)
plt.imshow(rotated_rgb)
plt.title("Rotated (45°)")
plt.axis("off")

# ---------------------------------------------------
# 6. Scaled Image
# ---------------------------------------------------
# Scale image by a factor of 1.5
scaled = cv.resize(img4, None, fx=1.5, fy=1.5)
scaled_rgb = cv.cvtColor(scaled, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 6)
plt.imshow(scaled_rgb)
plt.title("Scaled Image")
plt.axis("off")
```

```python
# ---------------------------------------------------
# 7. Sheared Image
# ---------------------------------------------------
# Apply horizontal shear transformation
M_shear = np.float32([[1, 0.5, 0],
                      [0, 1, 0]])
sheared = cv.warpAffine(img4, M_shear, (int(cols4 * 1.5), rows4))
sheared_rgb = cv.cvtColor(sheared, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 7)
plt.imshow(sheared_rgb)
plt.title("Sheared Image")
plt.axis("off")

# ---------------------------------------------------
# 8. Reflected Image
# ---------------------------------------------------
# Flip image horizontally
flipped = cv.flip(img4, 1)
flipped_rgb = cv.cvtColor(flipped, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 8)
plt.imshow(flipped_rgb)
plt.title("Reflected Image")
plt.axis("off")

# ---------------------------------------------------
# 9. Perspective Transformation
# ---------------------------------------------------
# Define source and destination points
pts1 = np.float32([
    [250, 350],
    [1450, 320],
    [300, 1250],
    [1400, 1200]
])

pts2 = np.float32([
    [0, 0],
    [1700, 0],
    [0, 1500],
    [1700, 1500]
])

# Compute perspective transformation matrix
M_persp = cv.getPerspectiveTransform(pts1, pts2)
perspective = cv.warpPerspective(img4, M_persp, (cols4, rows4))
perspective_rgb = cv.cvtColor(perspective, cv.COLOR_BGR2RGB)

plt.subplot(3, 3, 9)
plt.imshow(perspective_rgb)
plt.title("Perspective Transform")
plt.axis("off")
```

```python
# ---------------------------------------------------
# Adjust layout and display all plots
# ---------------------------------------------------
plt.tight_layout()
plt.show()
```